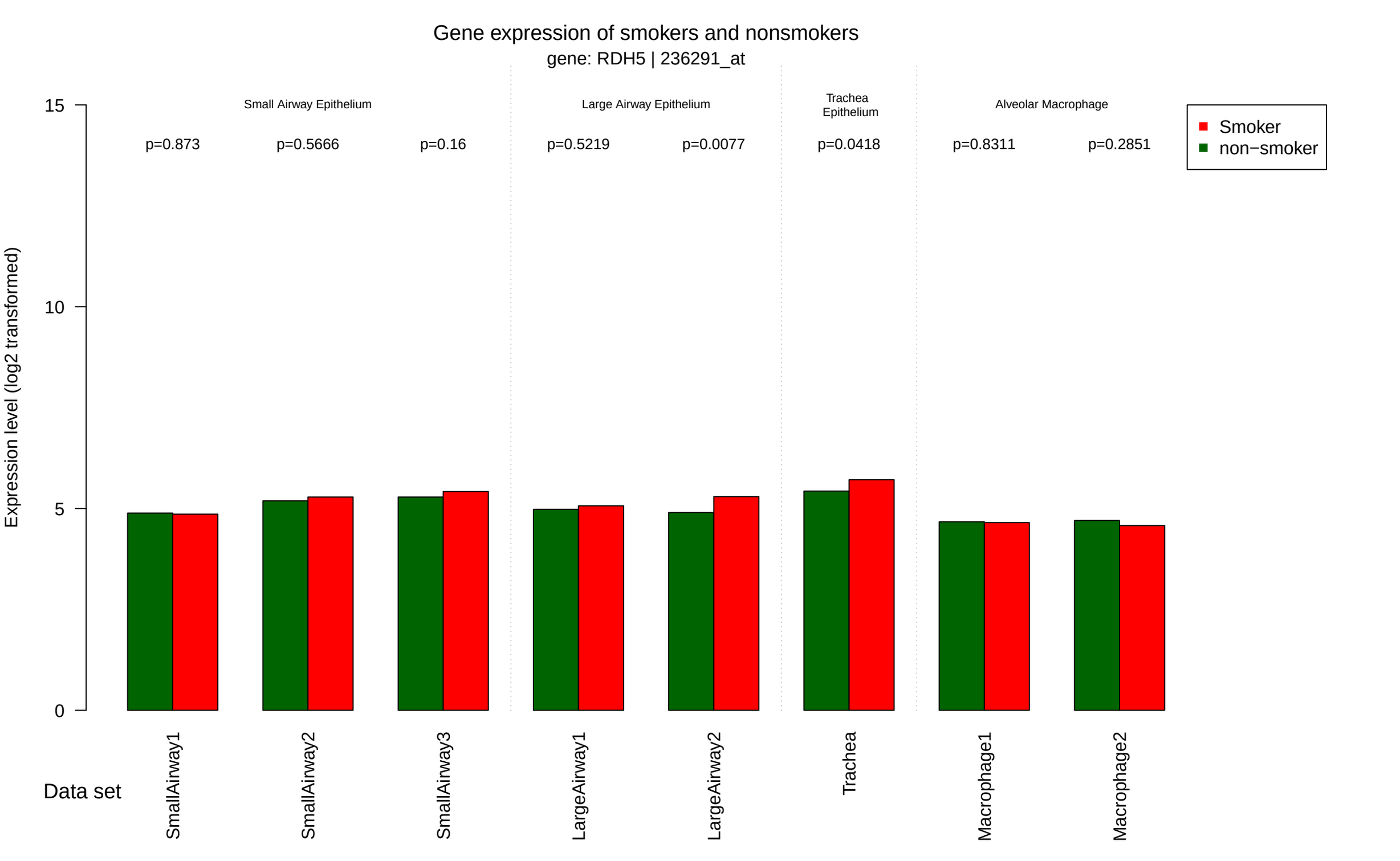
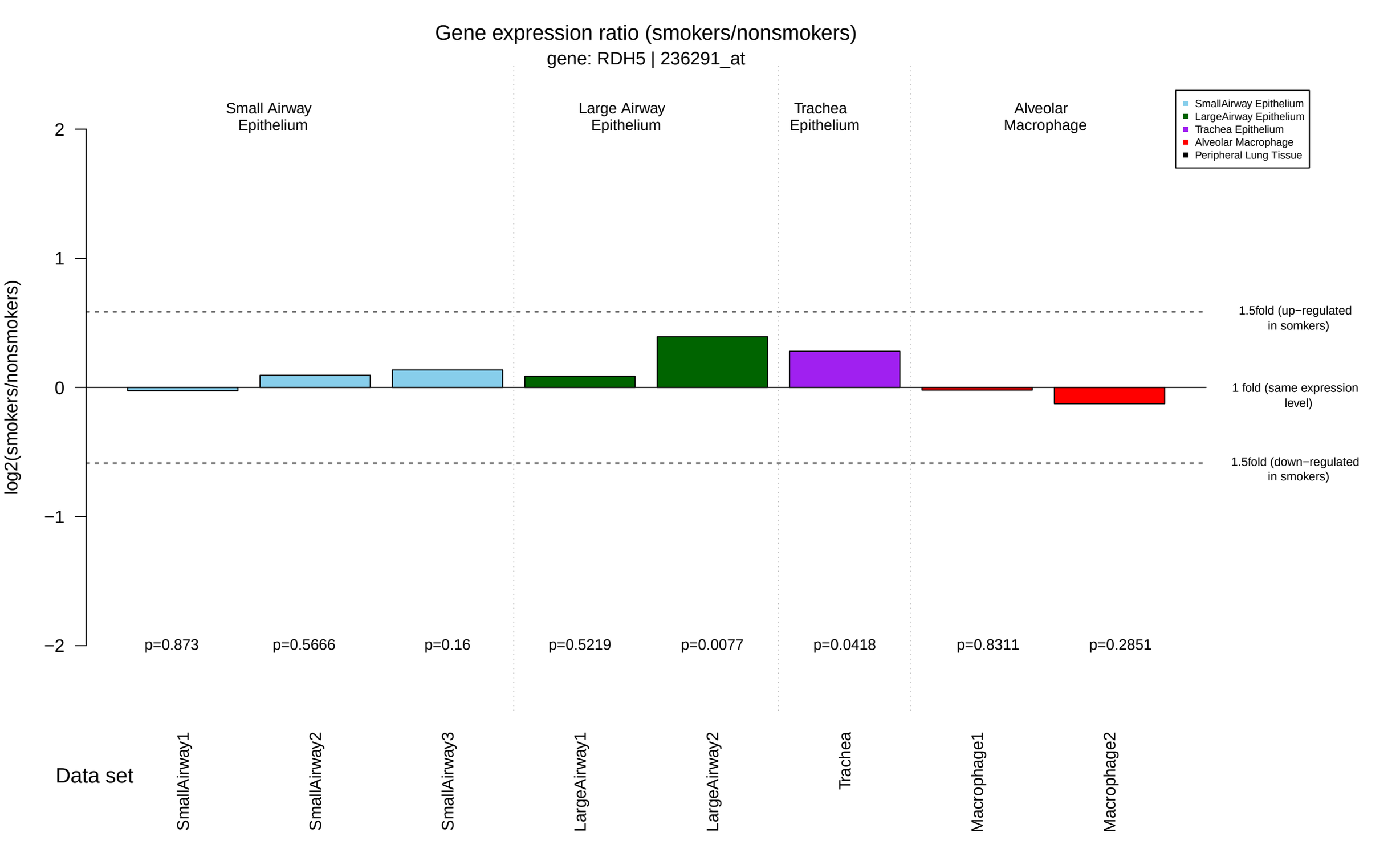
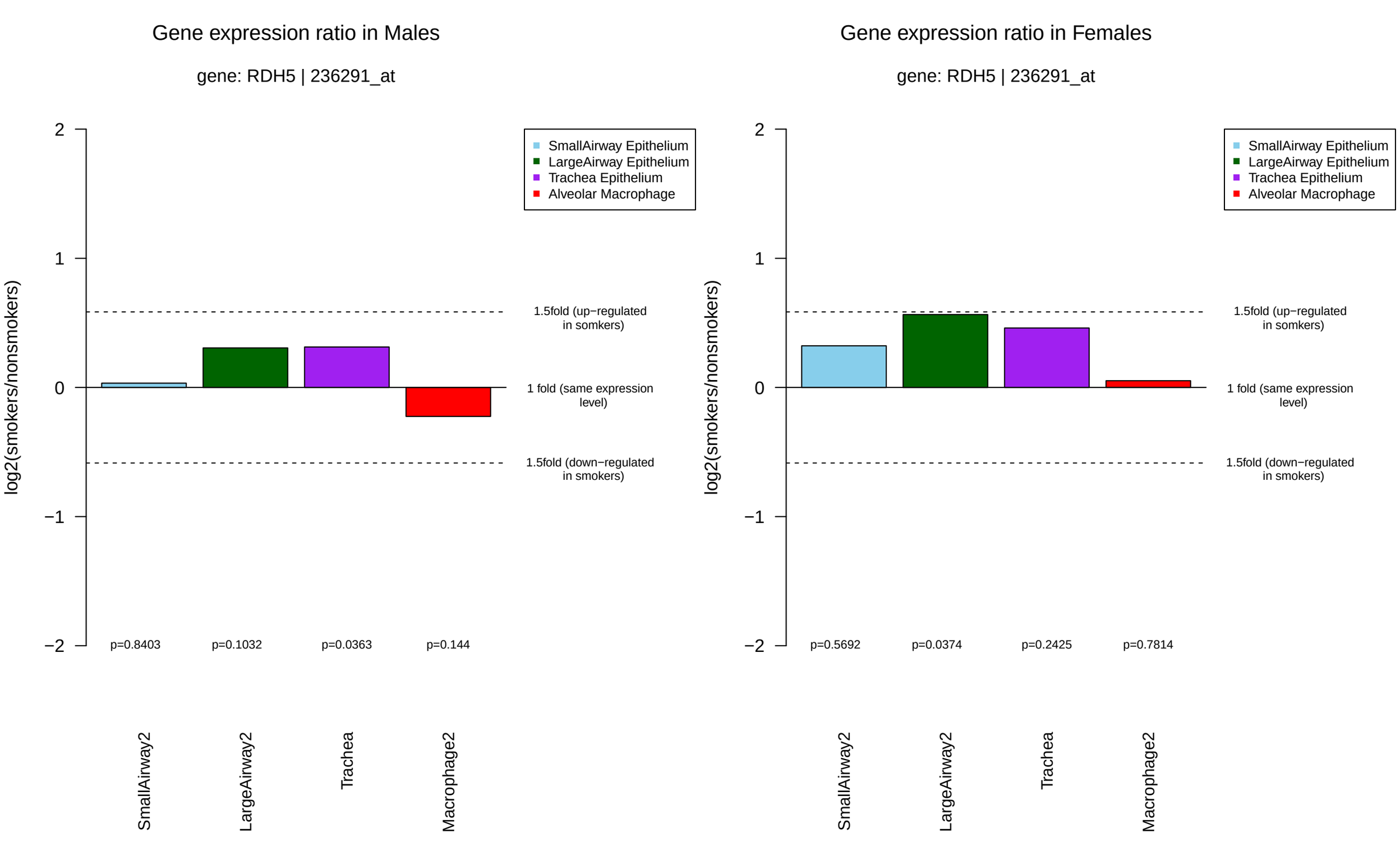
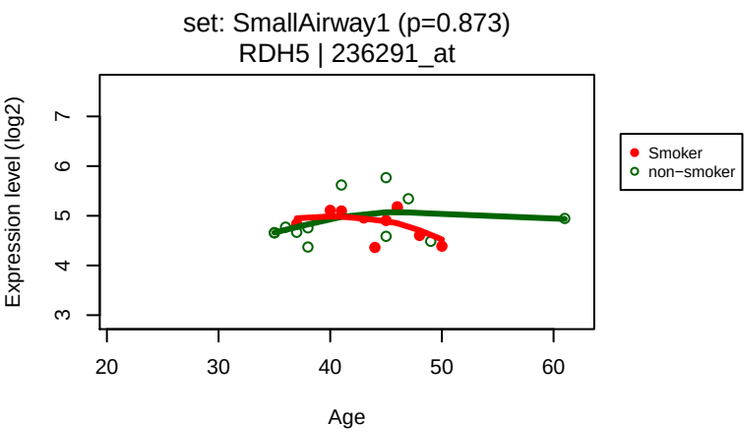
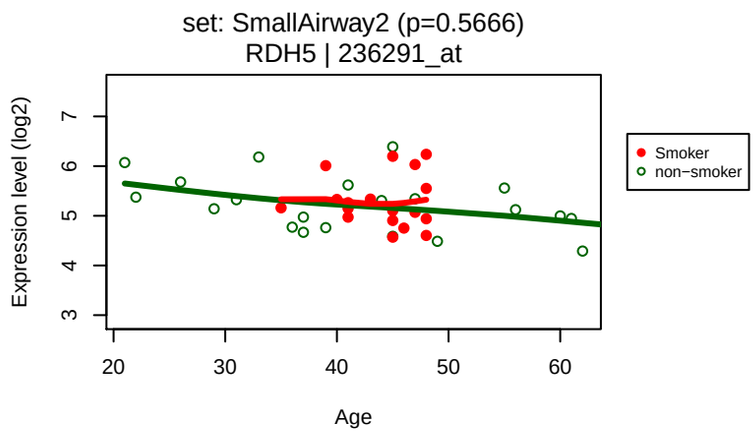
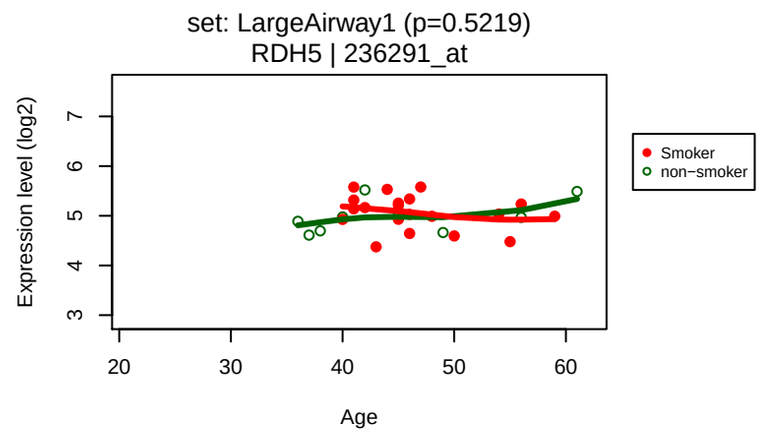

Gene expression of smokers and nonsmokers
gene: RDH5 | 236291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.873
p=0.5666
p=0.16
p=0.5219
p=0.0077
p=0.0418
p=0.8311
p=0.2851
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

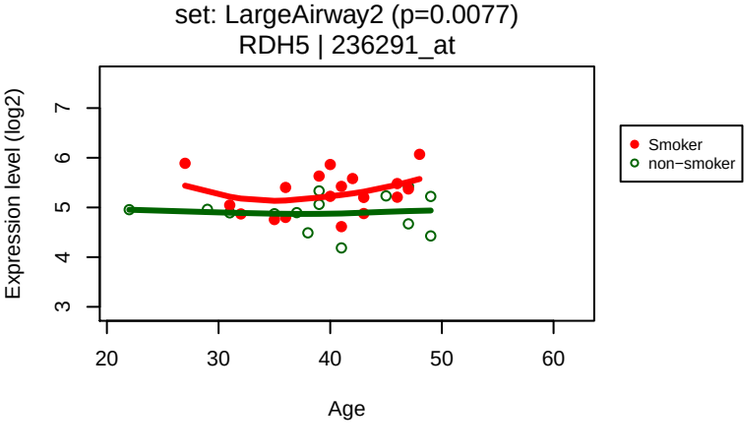

Gene expression ratio (smokers/nonsmokers)
gene: RDH5 | 236291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.873
p=0.5666
p=0.16
p=0.5219
p=0.0077
p=0.0418
p=0.8311
p=0.2851
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

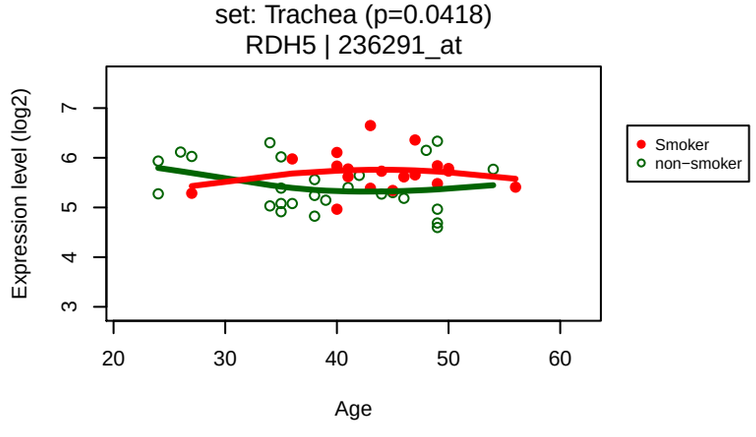

Gene expression ratio in Males
gene: RDH5 | 236291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8403
p=0.1032
p=0.0363
p=0.144
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RDH5 | 236291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5692
p=0.0374
p=0.2425
p=0.7814
SmallAirway2
LargeAirway2
Trachea
Macrophage2

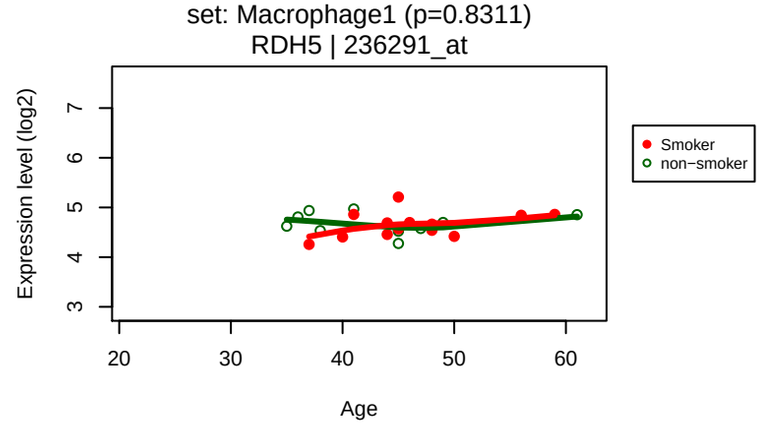

set: SmallAirway1 (p=0.873)
RDH5 | 236291_at
Expression level (log2)
Age
Smoker
non−smoker

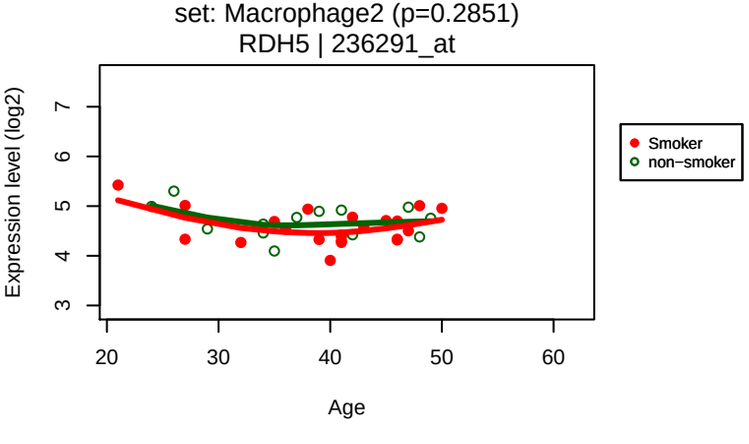

set: SmallAirway2 (p=0.5666)
RDH5 | 236291_at
Expression level (log2)
Age
Smoker
non−smoker

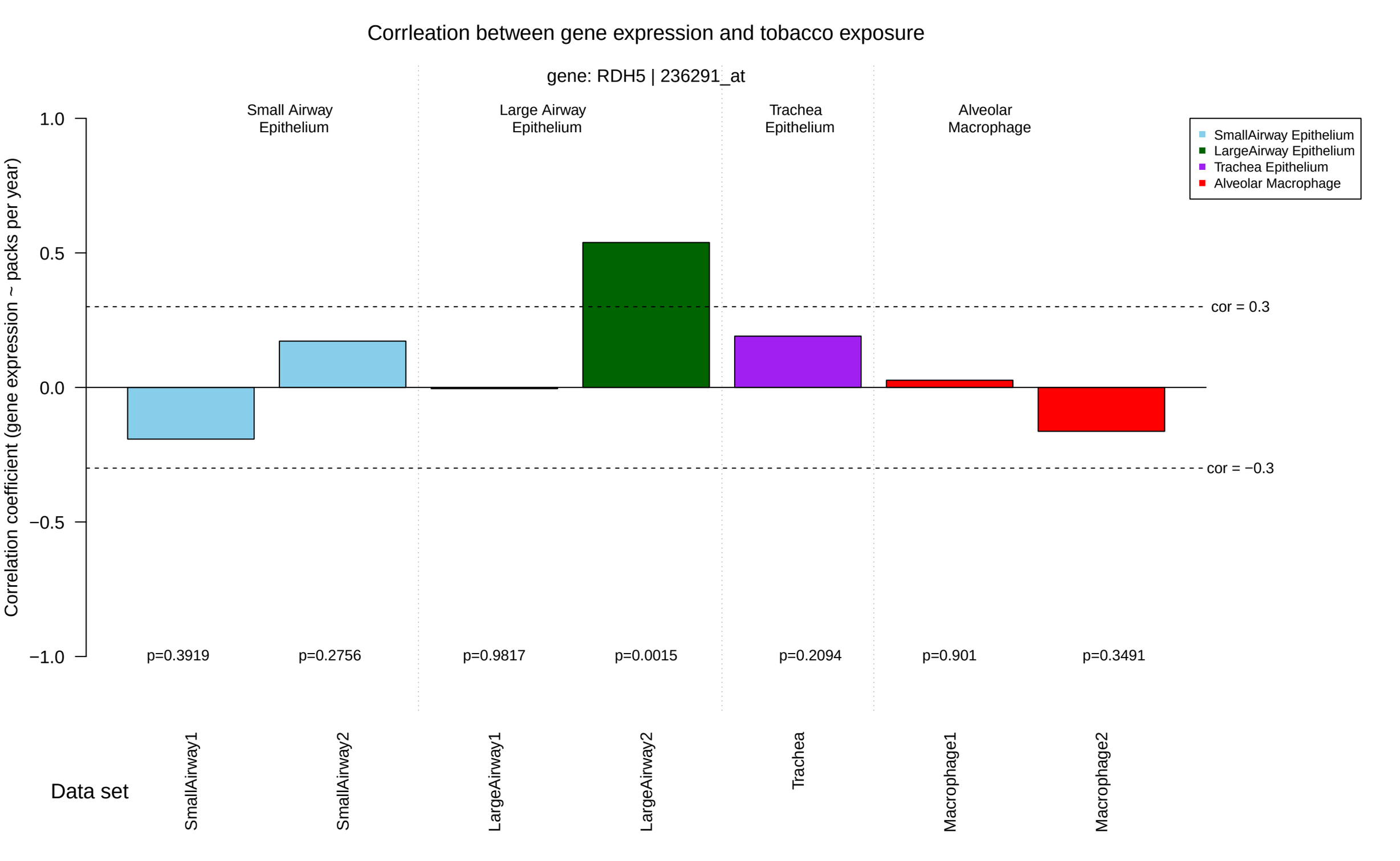

set: LargeAirway1 (p=0.5219)
RDH5 | 236291_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0077)
RDH5 | 236291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0418)
RDH5 | 236291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8311)
RDH5 | 236291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2851)
RDH5 | 236291_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RDH5 | 236291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3919
p=0.2756
p=0.9817
p=0.0015
p=0.2094
p=0.901
p=0.3491
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2